C. Flower City Fence

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

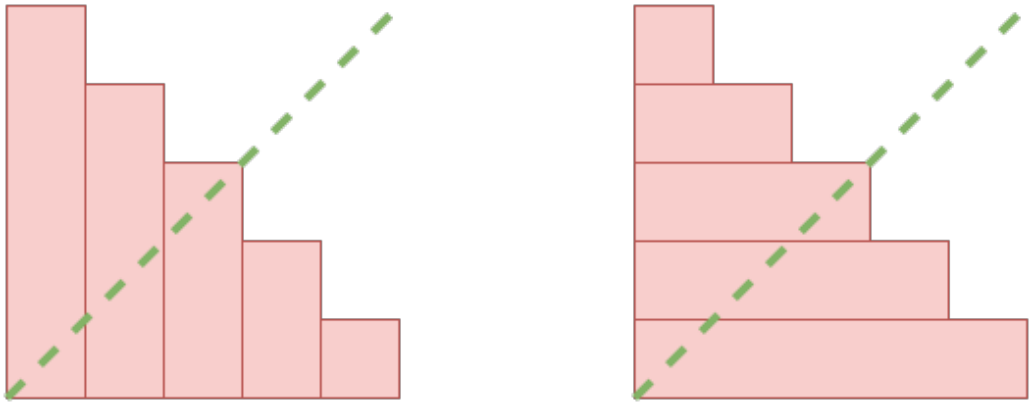
Rating: 1100 | Tags: binary search, data structures, implementation, sortings | https://codeforces.com/problemset/problem/1862/C

Anya lives in the Flower City. By order of the city mayor, she has to build a fence for herself.

The fence consists of n planks, each with a height of a_i meters. According to the order, the heights of the planks must **not increase**. In other words, it is true that $a_i \geq a_j$ for all i lt j.

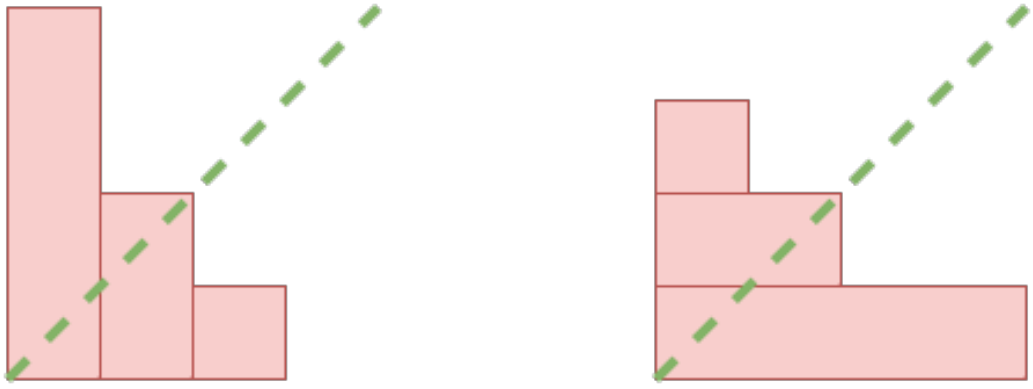
Anya became curious whether her fence is *symmetrical* with respect to the diagonal. In other words, will she get the same fence if she lays all the planks horizontally in the same order.

For example, for $n = 5$, $a = [5, 4, 3, 2, 1]$, the fence is *symmetrical*. Because if all the planks are laid horizontally, the fence will be [5, 4, 3, 2, 1], as shown in the diagram.



On the left is the fence [5, 4, 3, 2, 1], on the right is the same fence laid horizontally

But for $n = 3$, $a = [4, 2, 1]$, the fence is not *symmetrical*. Because if all the planks are laid horizontally, the fence will be [3, 2, 1, 1], as shown in the diagram.



On the left is the fence [4, 2, 1], on the right is the same fence laid horizontally

Help Anya and determine whether her fence is *symmetrical*.

Input

The first line of the input contains an integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The description of the test cases follows.

The first line of a test case contains a single integer n ($1 \le n \le 2 \cdot 10^5$) — the length of the fence.

The second line of a test case contains n integers $a_1 \ge a_2 \ge a_3 \ge \ldots \ge a_n$ ($1 \le a_i \le 10^9$) — the heights of the planks.

The sum of the values of n for all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output "YES" if the fence is *symmetrical*, otherwise output "NO".

You can output each letter in any case (lowercase or uppercase). For example, the strings "yEs", "yes", "Yes" and "YES" will be accepted as a positive answer.

Example

Input

```
7
5
5 4 3 2 1
3
3 1 1
3
4 2 1
1
2
5
5 3 3 1 1
5
5 5 5 3 3
2
6 1
```

Output

```
YES
YES
NO
NO
YES
YES
NO
```

Note

In the first and second test cases of the example, the fence is *symmetrical*.

In the third test case of the example, the fence is not *symmetrical*. If the planks are laid horizontally, the fence will be [3, 2, 1, 1].

In the fourth test case of the example, the fence is not *symmetrical*. If the planks are laid horizontally, the fence will be [1, 1].

In the fifth and sixth test cases of the example, the fence is *symmetrical*.

In the seventh test case of the example, the fence is not *symmetrical*. If the planks are laid horizontally, the fence will be [2, 1, 1, 1, 1, 1].